FACTORING A MAP OUT OF A K-FLAT COMPLEX THROUGH A K-FLAT QUASI-ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all clauses of the statement, including the preservation of flatness of the terms, and nothing is delegated.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules and the flatness of its terms. Same chapter, the definition of K-flatness and the criterion by acyclicity of total tensor products. Supplies the hypothesis on K^bullet and the property required of the auxiliary complex N^bullet.
- Triangulated structure of the homotopy category K(Mod(O)). Same chapter, distinguished triangles in the homotopy category and the long exact sequence of homotopy classes they induce. Supplies the argument by which the map a is replaced up to homotopy by a map into a quasi-isomorphic complex.
- Existence of K-flat resolutions with flat terms. Same chapter, the construction of a K-flat complex mapping quasi-isomorphically to a given complex, respecting flatness of the terms. Supplies the construction of N^bullet, the map c and their properties in the second part of the statement.

Source reference. Original lemma 014S: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3438–3438. Navigation pointer invoked at proof line 3438; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 06YS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Navigation pointer invoked at proof line 3444; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07A3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3081–3081. Navigation pointer invoked at proof line 3451; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 014A: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3473–3473. Navigation pointer invoked at proof line 3473; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G6C: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3478–3478. Navigation pointer invoked at proof line 3478; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $a : \mathcal{K}^\bullet \to \mathcal{L}^\bullet$ be a map of complexes of $\mathcal{O}$-modules. If $\mathcal{K}^\bullet$ is K-flat, then there exist a complex $\mathcal{N}^\bullet$ and maps of complexes $b : \mathcal{K}^\bullet \to \mathcal{N}^\bullet$ and $c : \mathcal{N}^\bullet \to \mathcal{L}^\bullet$ such that*

(1) *$\mathcal{N}^\bullet$ is K-flat,*
(2) *c is a quasi-isomorphism,*
(3) *a is homotopic to $c \circ b$.*

If the terms of $\mathcal{K}^\bullet$ are flat, then we may choose $\mathcal{N}^\bullet$, b, and c such that the same is true for $\mathcal{N}^\bullet$.

Proof. We will use that the homotopy category $K(\mathit{Mod}(\mathcal{O}))$ is a triangulated category, see Derived Categories, Proposition 014S. Choose a distinguished triangle $\mathcal{K}^\bullet \to \mathcal{L}^\bullet \to \mathcal{C}^\bullet \to \mathcal{K}^\bullet[1]$. Choose a quasi-isomorphism $\mathcal{M}^\bullet \to \mathcal{C}^\bullet$ with $\mathcal{M}^\bullet$ K-flat with flat terms, see Lemma 06YS. By the axioms of triangulated categories, we may fit the composition $\mathcal{M}^\bullet \to \mathcal{C}^\bullet \to \mathcal{K}^\bullet[1]$ into a distinguished triangle $\mathcal{K}^\bullet \to \mathcal{N}^\bullet \to \mathcal{M}^\bullet \to \mathcal{K}^\bullet[1]$. By Lemma 07A3 we see that $\mathcal{N}^\bullet$ is K-flat. Again using the axioms of triangulated categories, we can choose a map $\mathcal{N}^\bullet \to \mathcal{L}^\bullet$ fitting into the following morphism of distinguished triangles

$$\begin{array}{ccccccc} \mathcal{K}^\bullet & \longrightarrow & \mathcal{N}^\bullet & \longrightarrow & \mathcal{M}^\bullet & \longrightarrow & \mathcal{K}^\bullet[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathcal{K}^\bullet & \longrightarrow & \mathcal{L}^\bullet & \longrightarrow & \mathcal{C}^\bullet & \longrightarrow & \mathcal{K}^\bullet[1] \end{array}$$

Since two out of three of the arrows are quasi-isomorphisms, so is the third arrow $\mathcal{N}^\bullet \to \mathcal{L}^\bullet$ by the long exact sequences of cohomology associated to these distinguished triangles (or you can look at the image of this diagram in $D(\mathcal{O})$ and use Derived Categories, Lemma 014A if you like). This finishes the proof of (1), (2), and (3). To prove the final assertion, we may choose $\mathcal{N}^\bullet$ such that $\mathcal{N}^n \cong \mathcal{M}^n \oplus \mathcal{K}^n$, see Derived Categories, Lemma 0G6C. Hence we get the desired flatness if the terms of $\mathcal{K}^\bullet$ are flat. □